Constructing Vertex-Disjoint Paths in (*n*, *k*)-Star Graphs

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Abstract

This work develops a novel general routing algorithm, which is a routing function, for constructing a container of width n−1 between any pair of vertices in an (n, k)-star graph with connectivity n−1. Since the wide diameters in an (n, n−1)-star graph and an (n, 1)-star graph have been derived by Lin et al. (2004), this work only addresses an (n, k)-star with 2≤k≤n−2. The length of the longest container among all constructed containers serves as the upper bound of the wide diameter of an (n, k)-star graph. Moreover, this study also describes a lower bound of the wide diameter of an (n, k)-star graph with 2≤k≤⌊n/2⌋ and a lower bound of the wide diameter of a regular graph with connectivity ≥ 2. The measurement results demonstrate that the wide diameter of an (n, k)-star graph is its diameter plus 2 for 2≤k≤⌊n/2⌋, or between its diameter plus 1 and plus 2 for ⌊n/2⌋+1≤k≤n+2.

1 Introduction

An (n, k)-star graph ($S_{n,k}$) is a generalized version of an n-star graph (S_n), where $S_{n,n-1}$ and S_n are isomorphic and $S_{n,1}$ is a complete graph of n vertices (K_n) [5]. An S_n is a Caylay graph with regular and hierarchical structure, and had been recognized as an attractive alternative to an n-cube for constructing massively parallel computers [3], [4], [8], [9]. An S_n has significant advantages, such as low degree and small diameter [1], [7], [12]. Additionally, many advantageous properties, such as scalability, symmetry, and a high degree of fault tolerance, have been investigated [6], [10], [11], [14], [16], [17], [19], [20]. However, a rigorous restriction exists in that the number of vertices must satisfy $n!$. Thus, additional $(n-1)\times(n-1)!$ vertices are required to expand an S_{n-1} to an S_n. To eliminate this restriction, the new graph $S_{n,k}$ was developed [5]. An $S_{n,k}$ retains many of the attractive properties of an S_n such as vertex symmetry, hierarchical structure, maximal fault tolerance, and simple shortest-path routing. Only additional $(n-1)(n-1)!/(n-k)!$ vertices are utilized to expand an $S_{n-1,k-1}$ to an $S_{n,k}$. The topology of an $S_{n,k}$ is formally described in Section 2. In recent years, some basic properties of an $S_{n,k}$, such as diameter, connectivity, Hamiltonicity, spanning connectivity, and one-to-one routing, have been computed or derived [5], [14], [15], [18]. These measurement results demonstrate that an $S_{n,k}$ has very excellent topological properties.

A graph $G=(V, E)$ is an ordered pair in which vertex set $V(G)=\{v_1, v_2, \ldots, v_n\}$ and edge set $E(G)=\{e_1, e_2, \ldots, e_m\}$. Two vertices u and v are *adjacent* when they can be joined by an edge $e=(u, v)$. All vertices adjacent to a vertex are its *neighbors*. Given a distinct vertex sequence $\pi=v_0, v_1, \ldots, v_k$, where $v_i\in V(G)$, if v_{i-1} and v_i are adjacent for $i=1, 2, \ldots, k$, then π is called a (v_0, v_k)-*path* of length k. The *distance* from u to v, represented by $d(u, v)$, is the length of a shortest (u, v)-path for all (u, v)-paths in G. The *diameter* of G, represented by $d(G)$, is defined as the maximum of $d(u, v)$s for all pairs of distinct vertices u and v in G [2]. A graph is *connected* when at least one path exists between any two vertices. The *connectivity* of a graph is defined as the minimum number of vertices whose removal renders it disconnected or trivial. Let G be a graph whose connectivity is κ, and u and v are any two distinct vertices in the graph. According to Merger's theorem, at least κ vertex-disjoint paths exist between u and v in G [2]. A set of κ vertex-disjoint paths between u and v, denoted by $C_\kappa(u, v)$, is a (u, v)-*container* of width κ. The length of a $C_\kappa(u, v)$, denoted by $l(C_\kappa(u, v))$, is defined as the length of the longest path in $C_\kappa(u, v)$. A best-container between u and v, denoted by $C^*_\kappa(u, v)$, is the container among all $C_\kappa(u, v)$s, whose length is minimal. Let $d_\kappa(u, v)$ signify the κ-*wide distance* (or *wide distance*) from u to v; $d_\kappa(u, v)$ is then equal to $l(C^*_\kappa(u, v))$ [13]. The κ-*wide diameter* (or *wide diameter*) of G, denoted by $d_\kappa(G)$, is defined as the maximum of $d_\kappa(u, v)$s for all pairs of distinct vertices u and v in G. The wide diameter of G is very important in studies of reliability, fault tolerance, maximum parallelism, and minimum transmission delay in interconnection networks [13], [21].

Chiang and Chen in 1995 demonstrated that the connectivity of an $S_{n,k}$ is $n-1$ [5]. Therefore, by Merger's theorem, at least κ vertex-disjoint paths exist between distinct vertices u and v in an $S_{n,k}$.

This work first constructs a $C_{n-1}(u, v)$ between u and v in an $S_{n,k}$; the length of the longest container among all constructed containers serves as an upper bound of $d_{n-1}(S_{n,k})$. Then, this work demonstrates that the lower bound of $d_\kappa(G)$ for a regular graph with connectivity $\kappa\geq 2$ is its diameter plus 1. That is, $d_{n-1}(S_{n,k})$ is at least $d(S_{n,k})+1$ since an $S_{n,k}$ is regular and has a connectivity >1. Furthermore, constructing a best-container between a specific vertex pair u and v in an $S_{n,k}$ for $2\leq k\leq\lfloor n/2\rfloor$ acquired the finding that $d_{n-1}(S_{n,k})$ is at least $d(S_{n,k})+2$. Lin *et al.* (2004) demonstrated that $d_{n-1}(S_{n,k})=d(S_{n,k})+2$ [18]. They first constructed a $C_{n-1}(u, v)$ of maximum length $d(S_{n,k})+2$ between any vertex pair u and v in an $S_{n,k}$. Lin *et al.* then constructed a best-container of length $d(S_{n,k})+2$ between two specific nodes for each of $2\leq k\leq\lfloor n/2\rfloor$, $\lfloor n/2\rfloor+1\leq k\leq n+2$ if n is odd, and $\lfloor n/2\rfloor+1\leq k\leq n+2$ if n is even. Unfortunately, it has already been discovered that the best-containers constructed by Lin *et al.* for $\lfloor n/2\rfloor+1\leq k\leq n+2$ are not the best as they can be shortened by reconstructing longest paths of the containers. However, $d_{n-1}(S_{n,n-1})=d(S_{n,n-1})+2$ and $d_{n-1}(S_{n,1})=d(S_{n,1})+1$, computed by Lin *et al*. (2004), are correct [18]. Therefore, this work only focuses on an $S_{n,k}$ with $2\leq k\leq n-2$. This work demonstrates that $d_{n-1}(S_{n,k})$ is bounded above by $d(S_{n,k})+2$ and below by $d(S_{n,k})+1$. Furthermore, this work also determines that $d_{n-1}(S_{n,k})$ is $d(S_{n,k})+2$ for $2\leq k\leq\lfloor n/2\rfloor$.

The remainder of this paper is organized as follows. Section 2 formally describes the topology and properties of an $S_{n,k}$. Section 3 presents a path routing rule and discusses the length of constructed paths. Section 4 presents a general routing algorithm for $2\leq k\leq n-2$ and demonstrates that the constructed paths are vertex-disjoint and maximum length is $d(S_{n,k})+2$. Finally, conclusions are made in Section 5.

2 Background

An $S_{n,k}$ is an undirected-regular and vertex-symmetrical graph [5]. Parameters n and k, where $1\leq k\leq n-1$, are applied to determine the topological properties, such as number of vertices, connectivity, and diameter of an $S_{n,k}$. A vertex P can be represented as a label $p_1p_2...p_k$. The vertex set $V(S_{n,k})$ is defined as $\{p_1p_2...p_k \mid p_i\neq p_j$, for $i\neq j$ and p_i, p_j in $<n>\}$. Each element in $<n>$ is a *symbol*. Symbol $\in <k>$ ($\in <n>-<k>$) is an *internal* (*external*) symbol. Moreover, symbol p_i in label $p_1p_2...p_k$ is as follows: (1) *fixed* if $p_i=i$ (notably, position i is the *desired position* for symbol i); (2) *misplaced* if $p_i\neq i$ and p_i is an internal symbol; and, (3) *inner external* if p_i is an external symbol. A symbol not in the label $p_1p_2...p_k$ is *desired* if it is an internal symbol, or *outer external* if it is an external symbol. For instance, P=369482 in $S_{9,6}$, p_4=4 is a fixed symbol, and p_1=3, p_2=6 and p_6=2 are misplaced symbols, symbols 1 and 5 are desired symbols, and p_3=9 and p_5=8 are inner external symbols, and symbol 7 is an outer external symbol. Vertex P is adjacent to P_{p_i} and P_x, where

1. $P_{p_i}=p_ip_2...\ p_{i-1}p_1p_{i+1}...p_k$, for all $2\leq i\leq k$ (swapping p_1 with p_i), and
2. $P_x=xp_2...p_k$, for all x is a desired or outer external symbol (replacing p_1 with x).

Vertex ε=12...k is an *identity* vertex whose $p_i=i$ for all $1\leq i\leq k$. Since $S_{n,k}$ is vertex symmetry, it is the same from each of its vertices [5]; each vertex can be mapped to ε to investigate the basic properties of an $S_{n,k}$. This work attempts to identify $n-1$ vertex-disjoint paths from vertex P to ε by correcting each non-fixed symbol to a fixed symbol. All non-fixed symbols are represented as a *cycle representation*. That is, when $p_x=y$, $p_y=z$, and $p_z=x$, these tree non-fixed symbols can be presented as a cycle (x y z), where x (z) is the cycle *head* (*tail*), indicating that the desired position of a symbol is occupied by the next symbol in the cycle. Additionally, positions from k+1 to n, called *pseudo positions*, are not used in a label. The pseudo position of vertex P, denoted as P_{out} and vertex P can be relabeled as $P[P_{\text{out}}]$ = $p_1p_2...p_k[p_{k+1}...p_n]$. Consequently, a desired symbol and an outer external symbol can be considered misplaced and fixed, respectively, and each occupying a pseudo position. For instance, vertex P=369482 in $S_{9,6}$ can be extended as 369482[751], and represented by (3 9 1)(5 8)(2 6). Symbols 4 and 7 do not appear in the cycle representation and are regarded as fixed. Cycles (3 9 1) and (5 8) are external since an external symbol exists in each and symbols 1 and 5 are considered desired and misplaced. Cycle (2 6) is internal since all symbols are internal.

Let C_i be the i^{th} cycle of vertex P, and $|C_i|$ the number of symbols in C_i. Let m and e be the number of misplaced and inner external symbols, respectively. Then, in total, there are m+2e symbols in the cycle representation, implying that m+2e symbols are not at their desired positions. Moreover, if $p_1\neq 1$, C_1 is the cycle containing symbol 1 and is deemed a *primary cycle*.

Since the pseudo positions do not exist in a label, vertex P=369482 in $S_{9,6}$ can be represented by 369482[751] or 369482[715]. In other words, vertex P has two cycle representations: (3 9 1)(5 8)(2 6) and (3 9 5 8 1)(2 6). Notably, desired symbol 5 in cycle representation (3 9 5 8 1)(2 6) can be placed at pseudo position 9 rather than 8, such that external cycles (3 9 1) and (5 8) in cycle representation (3 9 1)(5 8)(2 6) can be merged into cycle (3 9 5 8 1). Merging all external cycles into one cycle occurs later using the proposed general routing algorithm (see Section 3). To distinguish between external cycles, the merged external cycle is called a *compound external cycle* (CEC).

For convenience, notations m_i, d_i, e_i, β_i and γ_i are used to denote the i^{th} misplaced, desired, inner external, fixed and outer external symbols, respectively. Furthermore, the i^{th} symbol can be denoted by s_i. Assuming that $C_iC_{i+1}...C_{i+j}=(d_i ... e_i)(d_{i+1} ... e_{i+1}) ... (d_{i+j} ... e_{i+j})$ are some external cycles of vertex P, the CEC of these cycles are denoted by $(C_iC_{i+1}...C_{i+j})_{\text{CEC}}=(d_i ... e_i\ d_{i+1} ... e_{i+1} ... d_{i+j} ... e_{i+j})_{\text{CEC}}$. The desired symbol d_{x+1} is regarded as the symbol residing at the pseudo position e_x. As mentioned, all external cycles in a given cycle representation can be combined and counted as 1.

Let c denote the number of cycles when all external cycles are combined as a CEC. Notably, distance $d(P, \varepsilon)$ is $c+m+2e$ if $p_1=1$ or $c+m+2e-2$ if $p_1\neq 1$ [5]. Obviously, $d(P, \varepsilon)$ is dominated by c, m and e. Moreover, [5] revealed that $d(S_{n,k})$ is $2k-1$ if $1\leq k\leq \lfloor n/2\rfloor$ or $k+\lfloor (n-1)/2\rfloor$ if $\lfloor n/2\rfloor\leq k\leq n-1$. Notice that an internal cycle contains at least two misplaced symbols and an external cycle contains exactly one desired symbol collocating one inner external symbol. Therefore, other misplaced symbols, called *redundant* symbols, in a cycle have no effect on parameter c. Let m' be the number of redundant symbols and since there are x internal cycles in the cycle representation, $m'=m-2x$, implying that m' is even (or odd) if m is even (or odd). Consequently, $c=(m-m')/2+1$ if $e\neq 0$ or $(m-m')/2$ if $e=0$.

3 Simple Routing

The two simple routing rules used to construct a path from P to ε in an $S_{n,k}$ are first described in this section. Next, a symbol sequence is defined and utilized to construct a shortest path. Constructed path length can be derived as the frequency of applied rules. Moreover, $d(P, \varepsilon) \leq d(S_{n,k})$ and can be represented as $d(S_{n,k})-\Delta$, where Δ is a nonnegative integer. This work derived the formulae for Δ for determining the $l(C_{n-1}(P, \varepsilon))$.

3.1 Path Routing

To construct a path π from P to ε in an $S_{n,k}$, all misplaced, desired and inner external symbols of P should be corrected one by one to transform the label P into label ε. The following routing rules are applied for correcting symbol s, and each rule can join 1 or 2 edges (vertices) in a path [5].

R1: If symbol s is at position 1, then

- If s is an internal symbol, move it to its desired position. (The symbol residing at position s is automatically moved to position 1 when s is corrected.)
- If s is an external symbol, exchange s with a desired symbol.

R2: If symbol s is not at position 1, then

- Exchange s with the symbol in position 1 and then apply rule R1.

One edge contributed to path π if R1 is applied and two edges contributed to path π if R2 is applied. Clearly, all symbols can be corrected in different orders such that different paths from P to ε can be constructed. The following lemma presents a condition in which two constructed paths are vertex-disjoint.

Lemma 1. Assuming π_1 and π_2 are two distinct paths from P to ε and constructed by R1 and/or R2, corrections of symbols a and i (b and j) are the first (last) corrections in π_1 and π_2, respectively. If the correction order of b is prior to a in π_2 then π_1 and π_2 are vertex-disjoint except P and ε.

Proof. Let $n_t(\pi)$ be the t^{th} vertex in path π, where $n_0(\pi)=P$. First, $n_1(\pi_1)$ is disjoint to each vertex in π_2 as $n_1(\pi_1)$ and $n_1(\pi_2)$ are distinct neighbors of P, and symbol i in labels of the other vertices in π_2 has already been corrected but $n_1(\pi_1)$ has not yet. Second, each vertex $n_x(\pi_1)$, $x\geq 2$, has $p_a=a$ and $p_b\neq b$. Since the correction order of b is before that for a in π_2, b is corrected before a such that all vertices in π_2 do not have $p_a=a$ and $p_b\neq b$. Therefore, π_1 and π_2 are vertex-disjoint.

Assuming vertex P's cycle representation is $C_1C_2...C_c$, let $C_i=(s_{i,1}\ s_{i,2} ... s_{i,k_i})$ and $k_i=|C_i|$. Symbol $s_{i,x}$ is then considered the x^{th} symbol in C_i, thus, $s_{i,1}$ is the head and s_{i,k_i} is the tail of C_i. Particularly, if C_1 is a primary cycle, then regularize symbol 1 as its tail s_{1,k_1} and p_1 as the head $s_{1,1}$ by rotating C_1. A *symbol sequence* is defined as $\sigma=\langle C_1, C_2, ...,C_c\rangle =\langle s_{1,1}, s_{1,2}, ..., s_{1,k_1}, s_{2,1}, s_{2,2}, ...,s_{2,k_2}, ..., s_{c,k_c}\rangle$. For convenience, σ is also renumbered as $\langle s_1, s_2, ...,s_{m+2e}\rangle$, where $k_1+k_2+...+k_c=m+2e$. These two expressions are used alternately throughout the remainder of this work.

If path π is constructed in the order σ, each of $m+2e$ symbols should be corrected by applying R1 or R2 to construct π. If $p_1=1$, all c heads are not at position 1 when these heads turn to be corrected and R2 is applied. Conversely, each of remaining $m+2e-c$ symbols in σ can be corrected by applying R1 since the symbol can be automatically located at position 1 as a result of correcting its previous symbol in σ. Thus, $2c+(m+2e-c)=c+m+2e$ edges are contributed to the constructed path π. If $p_1\neq 1$, all c heads except $s_{1,1}$ are not at position 1 and the correction of s_{1,k_1}

is no longer required as position 1 is its desired position and $2(c-1)+(m+2e-c)= c+m+2e-2$ edges would be contributed to the constructed path π. Let $l(\pi)$ denote the length of path π. Therefore, path π is a shortest path and $l(\pi)$ is equal to $d(P, \varepsilon)$ [5].

3.2 Length analysis

As $d(P, \varepsilon)\le d(S_{n,k})$, $d(P, \varepsilon)$ can be represented as $d(S_{n,k})-\Delta$, where $\Delta=d(S_{n,k})-d(P, \varepsilon)$ is called a *difference*. For vertex P in a specific $S_{n,k}$, Δ is only related to $d(P, \varepsilon)$ since $d(S_{n,k})$ is invariant. As the fixed symbols and outer external symbols do not require correction when constructing a shortest path and increasing the number of redundant symbols reduces the number of cycles, Δ is dominated by the number of misplaced, redundant, fixed, and outer external symbols of P.

For vertex P in a specific $S_{n,k}$, Δ is also denoted as Δ_1 if $2\le k\le\lfloor n/2\rfloor$, Δ_2 if $\lfloor n/2\rfloor+1\le k\le n-2$ and n is even, or Δ_3 if $\lfloor n/2\rfloor+1\le k\le n-2$ and n is odd. Assuming that P contains f fixed symbols other than symbol 1, and e' outer external symbols, then Δ_1, Δ_2, and Δ_3 are derived in the following lemmas for further computation.

Lemma 2. When $e>0$, $\Delta_1=(m+m'+4f)/2$, $\Delta_2=(m'+e'+3f-1)/2$ if $p_1=1$ or $\Delta_2=(m'+e'+3f)/2$ if $p_1\ne 1$, and $\Delta_3=(m'+e'+3f)/2$ if $p_1=1$ or $\Delta_3=(m'+e'+3f+1)/2$ if $p_1\ne 1$.

Proof. Recall that $d(P, \varepsilon) = c+m+2e$ ($d(P, \varepsilon) = c+m+2e-2$) if $p_1=1$ ($p_1\ne 1$) and $d(S_{n,k}) = 2k-1$ ($d(S_{n,k}) = k+\lfloor(n-1)/2\rfloor$) if $2\le k\le\lfloor n/2\rfloor$ ($\lfloor n/2\rfloor+1\le k\le n-2$) [5]. Assuming $p_1=1$ (or $p_1\ne 1$) and $2\le k\le\lfloor n/2\rfloor$, then $\Delta_1=d(S_{n,k})-d(P, \varepsilon) = (2k-1)-(c+m+2e) = (m+m'+4f)/2$ (or $(2k-1)-(c+m+2e-2) = (m+m'+4f)/2$), since $k= m+e+f+1$ (or $k=m+e+f$) and $c=(m-m')/2+1$. For $\lfloor n/2\rfloor+1\le k\le n-2$, because $n=k+e+e'$, Δ_2 and Δ_3 can be similarly obtained.

Lemma 3. When $e=0$, $\Delta_1=(m+m'+4f)/2+1$, $\Delta_2=(m'+e'+3f-1)/2+1$ if $p_1=1$ or $\Delta_2=(m'+e'+3f)/2+1$ if $p_1\ne 1$, and $\Delta_3=(m'+e'+3f)/2+1$ if $p_1=1$ or $\Delta_3=(m'+e'+3f+1)/2+1$ if $p_1\ne 1$.

Proof. Given $c=(m-m')/2$, this proof is similar to that for Lemma 2.

Since $d(S_{n,k})$ and $d(P, \varepsilon)$ are positive integers and $d(P, \varepsilon)\le d(S_{n,k})$, Δ is a nonnegative integer. By Lemmas 2 and 3, the following corollary is obtained easily.

Corollary 4. Assuming $\lfloor n/2\rfloor+1\le k\le n-2$ and n is even (odd), $m'+e'+3f$ is odd (even) if $p_1=1$ or $m'+e'+3f$ is even (odd) if $p_1\ne 1$.

4 General Routing Algorithm

This section presents the proposed general routing algorithm for constructing $n-1$ vertex-disjoint paths from a source vertex P to the destination vertex ε for $2\le k\le n-2$. The proposed general routing algorithm is based on and modified from [6], which developed an efficient algorithm for constructing $n-1$ vertex-disjoint paths between any two vertices in S_n. First, the proposed general routing algorithm is introduced and the lengths of the constructed $n-1$ vertex-disjoint paths are also computed. Hence, based on the lengths of the constructed $n-1$ vertex-disjoint paths, a upper bound of $d_{n-1}(S_{n,k})$ is determined. Finally, a lower bound of $d_{n-1}(S_{n,k})$ is also derived.

4.1 Algorithm

The general routing algorithm is classified into α-function, β-function and γ-function for $p_1=1$ and $p_1\ne 1$. These three functions are described as follows.

α-function:

Let $\sigma(s_\alpha)$ be the symbol sequence that performs $\alpha-1$ times a left circular shift on σ such that s_α is the first symbol and $s_{\alpha-1}$ is the last symbol of $\sigma(s_\alpha)$, where $1\le\alpha\le m+2e$. A path $\pi(s_\alpha)$ is generated by correcting symbols according to the order of $\sigma(s_\alpha)$. If $p_1=1$, each path is constructed by a sequence $\sigma(s_\alpha)$. If $p_1\ne 1$, each path is constructed as follows:

(1) For all heads $s_\alpha=s_{i,1}$, $1\le i\le c$: each path is constructed by $\sigma(s_\alpha)$.

(2) For all $s_\alpha\ne s_{i,1}$, they are not a head and in C_i, for $2\le i\le c$: Let $C_i=(S_{il}\ S_{ir})$, where $S_{il}=s_{i,1}\ s_{i,2}\ ...,s_{\alpha-1}$ and $S_{ir}=s_\alpha\ s_{\alpha+1}\ ...\ s_{i,k_i}$. The sequence is modified as $\sigma(s_\alpha)=\langle s_\alpha, ..., s_{\alpha-1}\rangle=\langle S_{ir}, C_{i+1}, ..., C_c, s_{1,k_1}, s_{1,1}, s_{1,2}, ..., s_{1,k_1-1}, C_2, ..., C_{i-1}, S_{il}\rangle$, where C_1 is rotated as $(s_{1,k_1}\ s_{1,1}\ s_{1,2}\ ...\ s_{1,k_1-1})$.

(3) All s_α do not form a head and belong to C_1. Notably, paths $\pi(s_{1,2})$ and $\pi(s_{1,1})$ are identical if $p_1\ne 1$. Because $\sigma(s_{1,1})=\langle s_{1,1}, s_{1,2}, ..., s_{c,k_c}\rangle$ and $\sigma(s_{1,2})=\langle s_{1,2}, s_{1,3}, ..., s_{c,k_c}, s_{1,1}\rangle$, correcting $s_{1,1}$ for $\pi(s_{1,1})$ and $s_{1,2}$ for $\pi(s_{1,2})$ are the same; that is, take $s_{1,1}$ to its desired position and $s_{1,2}$ to position 1. In other words, sequences $\sigma(s_{1,1})$ and $\sigma(s_{1,2})$ construct the same path, implying that only $|C_1|\ge 3$ should be considered here.

 (3.1) If $s_{1,1}$ is an external symbol and s_{1,k_1} is a desired symbol, construct paths by $\langle s_{1,k_1}, C_2, ..., C_c, s_{1,3}, ..., s_{1,k_1-1}, s_{1,1}, s_{1,2}\rangle$ for $\pi(s_{1,k_1})$, $\langle s_{1,3}, ..., s_{1,k_1}, C_2, ..., C_c, s_{1,2}, s_{1,1}\rangle$ for $\pi(s_{1,3})$, and $\langle s_\alpha, s_{\alpha+1}, ..., s_{\alpha-1}\rangle$ for $\pi(s_\alpha)$, where $s_\alpha\ne s_{1,k_1}$ and $s_\alpha\ne s_{1,3}$.

(3.2) Otherwise, construct paths by $<s_{1,k_1}, C_2, ..., C_c, s_{1,2}, ..., s_{1,k_1-1}, s_{1,1}>$ for $\pi(s_{1,k_1})$ and $<s_\alpha, s_{\alpha+1}, ..., s_{\alpha-1}>$ for $\pi(s_\alpha)$, where $s_\alpha \neq s_{1,k_1}$.

Lemma 5. There are $m+2e$ (or $m+2e-1$) vertex-disjoint paths constructed by the α-function if $p_1=1$ (if $p_1\neq 1$).

Proof. If $p_1=1$, each path is constructed via sequence $\sigma(s_\alpha)$, where $1\leq\alpha\leq m+2e$. There are $m+2e$ paths that can be constructed. Let $\sigma(s_{\alpha'})$ be the symbol sequence that performs some times a left circular shift on $\sigma(s_\alpha)$. Because $s_{\alpha-1}$ is prior to s_α in $\sigma(s_{\alpha'})$, by Lemma 1, $\pi(s_\alpha)$ is vertex-disjoint to $\pi(s_{\alpha'})$. Therefore, all constructed paths are vertex-disjoint.

If $p_1\neq 1$, since $\pi(s_{1,1})$ and $\pi(s_{1,2})$ are identical, only $m+2e-1$ paths can be constructed. The vertex-disjoint property of these $m+2e-1$ paths is described as follows:

(1) For all heads $s_{i,1}$, $1\leq i\leq c$, there are c paths constructed by $\sigma(s_{i,1})$. The sequence $\sigma(s_{i,1})$ is the same as $p_1=1$ and the vertex-disjoint property can be proved similarly.

(2) For all s_α, they are not a head and all belong to C_i, $2\leq i\leq c$. There are $m+2e-c-|C_1|+1$ paths constructed. Each path is constructed via a modified $\sigma(s_\alpha)$, and the vertex-disjoint property can be similarly proved as $p_1=1$. Furthermore, the correcting order of $s_{\alpha-1}$ is prior to that of s_α in each path constructed by (1) since s_α is not a head, thereby implying that $s_{\alpha-1}$ and s_α are in the same cycle. Note that $s_{\alpha-1}$ may be a head. Therefore, by Lemma 1, each path $\pi(s_\alpha)$ is vertex-disjoint to each path constructed by (1).

(3) For all s_α, they are not a head and belong to C_1. There are $|C_1|-2$ paths constructed.

(3.1) If $s_{1,1}$ is an external symbol and s_{1,k_1} is a desired symbol. After the first correction when constructing $\pi(s_{1,k_1})$, $p_1=s_{1,1}$ is replaced with the desired symbol $s_{1,k_1}=1$. Therefore, no vertices in $\pi(s_{1,k_1})$ have the external symbol $s_{1,1}$. When constructing $\pi(s_{1,3})$, $s_{1,1}$ is corrected during the last step and all vertices in $\pi(s_{1,3})$ have the external symbol $s_{1,1}$. Therefore, $\pi(s_{1,k_1})$ and $\pi(s_{1,3})$ are vertex-disjoint. By Lemma 1, all $\pi(s_\alpha)$s other than $\pi(s_{1,k_1})$ and $\pi(s_{1,3})$ are vertex-disjoint to each other and are vertex-disjoint to $\pi(s_{1,k_1})$ and $\pi(s_{1,3})$. Consequently, all constructed paths are vertex-disjoint. Moreover, $\pi(s_{1,k_1})$ and $\pi(s_{1,3})$ are vertex-disjoint to each path constructed by (1) since the order of $s_{1,2}$ ($s_{1,1}$) is prior to that of s_{1,k_1} ($s_{1,3}$) in $\sigma(s_{i,1})$, $1\leq i\leq c$. Additionally, the order of $s_{\alpha-1}$ is prior to that of s_α in $\sigma(s_{i,1})$ since s_α is not a head and $s_{\alpha-1}$ and s_α are both in C_1. Therefore, by Lemma 1, each path $\pi(s_\alpha)$ is vertex-disjoint to each path constructed by (1). Consequently, all paths constructed by (3.1) are vertex-disjoint to each path constructed by (1). Similarly, all paths constructed by (3.1) are vertex-disjoint to each path constructed by (2).

(3.2) Otherwise, the proof is similar to that for (3.1) and all constructed paths are all vertex-disjoint to each other and vertex-disjoint to each path constructed by (1) and (2).

Lemma 6. The lengths of paths constructed by the α-function are as follows:

(1) If $p_1=1$, the $m+2e$ paths are at most $d(S_{n,k})$ long.

(2) If $p_1\neq 1$, c paths are at most $d(S_{n,k})$ long, and $m+2e-c-1$ paths are at most $d(S_{n,k})+1$ ($d(S_{n,k})+2$) long when $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is odd (when $2\leq k\leq\lfloor n/2\rfloor$ or $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is even).

Proof. If $p_1=1$, assuming that s_α is a symbol of C_i, $1\leq i\leq c$, R2 is applied to correct s_α, since s_α is not at position 1. Suppose $s_{\alpha+1}$ is the symbol in C_i and follows s_α, then $s_{\alpha+1}$ can be automatically moved to position 1 once s_α is corrected. If $s_{\alpha+1}$ is to be corrected next, R1 can be applied. Other symbols except s_α in the primary cycle are the same. Moreover, R2 is also applied to correct the head of each cycle other than C_i because such head is also not at the position 1 when it is its turn to be corrected. In total, s_α and $c-1$ heads apply R2 and the remaining $m+2e-c$ symbols apply R1. Therefore, $2c+(m+2e-c) = c+m+2e = d(P, \varepsilon)$ edges would contribute to the constructed path. Therefore, $l(\pi(s_\alpha))$ is $d(P, \varepsilon)\leq d(S_{n,k})$.

If $p_1\neq 1$, Δ_1, $\Delta_2\geq 0$ and $\Delta_3\geq 1$ can be obtained by Lemma 2, Lemma 3 and Corollary 4. The lengths of the constructed paths are computed as follows:

(1) For all paths $\pi(s_{i,1})$, $1\leq i\leq c$. The path construction is similar to that for $p_1=1$. All c heads other than $s_{1,1}$ are not at position 1 and R2 should be applied. However, correcting s_{1,k_1} is no longer required since position 1 is its desired position. Thus, $2(c-1)+(m+2e-c) = d(P, \varepsilon)$ edges would contribute to the constructed path. Therefore, $l(\pi(s_{i,1}))$ is $d(P, \varepsilon)\leq d(S_{n,k})$.

(2) For all paths $\pi(s_\alpha)$, where s_α is not a head and in C_i, $2\leq i\leq c$. Recall that $\sigma(s_\alpha)$ is $<s_\alpha, s_{\alpha+1}, ..., s_{i,k_i}, C_{i+1}, ..., C_c, s_{1,k_1}, s_{1,1}, s_{1,2}, ..., s_{1,k_1-1}, C_2, ..., C_{i-1}, s_{i,1}, s_{i,2}, ..., s_{\alpha-1}>$. s_{1,k_1} (symbol 1) is corrected prior to other symbols in C_1. Initially, $p_1\neq s_\alpha$, and R2 is applied to correct s_α. Next, symbols $s_{\alpha+1}, ..., s_{i,k_i}, s_{i,1}\ s_{i,2}\ ..., s_{\alpha-1}$ are

automatically moved to position 1, after their previous symbols are corrected, and R1 is applied. Additionally, s_{i,k_i} is the symbol before $s_{i,1}$ in the rotated C_i. Once s_{i,k_i} is corrected, $s_{i,1}$ can be automatically moved to position 1, and the head of C_{i+1} is now to be corrected and R2 must be applied. Indeed, the heads of C_{i+2}, ..., C_c, C_2, ..., C_{i-1}, are the same. Moreover, $s_{i,1}$ will return to position 1 automatically when each cycle is corrected. Hence, when s_{1,k_1} after C_c is to be corrected, $s_{i,1}$ should be exchanged with s_{1,k_1}, and then $s_{1,1}$ is due for correction, and s_{1,k_1} is now at position 1. Consequently, R2 should be applied to correct $s_{1,1}$. For the remaining symbols, R1 is applied. In total, c symbols apply R2, and $s_{i,1}$ and s_{1,k_1} are exchanged once. Other $m+2e-c-1$ symbols apply R1. Thus, $2c+1+(m+2e-c-1)=c+m+2e=d(P, \varepsilon)+2$ edges would contribute to the constructed path. Therefore, $l(\pi(s_\alpha))$ is $d(P, \varepsilon)+2=d(S_{n,k})-\Delta_3+2\le d(S_{n,k})+1$ if $\lfloor n/2\rfloor+1\le k\le n-2$ and n is odd, or $d(P, \varepsilon)+2=d(S_{n,k})-\Delta_j+2\le d(S_{n,k})+2, j\in\{1, 2\}$ otherwise.

(3) All s_α are not a head and belong to C_1. Since the construction is similar to that for (2), the lengths of the constructed paths are at most $d(S_{n,k})+1$ if $\lfloor n/2\rfloor+1\le k\le n-2$ and n is odd or $d(S_{n,k})+2$ otherwise.

β-function:

Let β_i be a fixed symbol in vertex P, excluding symbol 1, where $1\le i\le f$. The β-function is designed to move one of P's fixed symbols, β_i, to position 1; this process hops one step to P's neighbor denoted by P_{β_i}, and then keeps β_i away from its desired position until the other symbols have been corrected. The ith path constructed by the β-function is denoted by $\pi(\beta_i)$. Notably, symbol 1 cannot be used by β-function even though it is fixed ($p_1=1$). Clearly, no $\pi(\beta_i)$s can be constructed if $f=0$. A $\pi(\beta_i)$ is constructed as follows.

Step 1. Move β_i to position 1. This process hops one step to P_{β_i} and its primary cycle is $(\beta_i\ 1)$ if $p_1=1$ (or $(\beta_i\ C_1)$ if $p_1\ne1$).

Step 2. If $p_1=1$, construct a path from P_{β_i} to ε using the sequence $<C_1, C_2, ..., C_c, \beta_i.>$. If $p_1\ne1$, construct a path from P_{β_i} to ε using the sequence $<C_2, ..., C_c, s_{1,k_1}, s_{1,1}, s_{1,2}, ..., s_{1,k_1-1}, \beta_i>$.

Lemma 7. There are $k-m-e-1$ ($k-m-e$) vertex-disjoint paths constructed by the β-function if $p_1=1$ ($p_1\ne1$). Each $\pi(\beta_i)$ is vertex-disjoint to all $\pi(s_\alpha)$s.

Proof. Obviously, f fixed symbols are used to construct path(s) via the β-function. By definition, $f=k-m-e-1$ ($k-m-e$) if $p_1=1$ ($p_1\ne1$). Clearly, each β_i is distinct and all $\pi(\beta_i)$s are vertex-disjoint to each other. Since all vertices in a constructed path other than $\pi(\beta_i)$ have β_i at their desired position, each $\pi(\beta_i)$ is vertex-disjoint to all $\pi(s_\alpha)$s.

Lemma 8. The lengths of paths constructed by the β-function are as follows.

(1) If $p_1=1$, the lengths of paths are at most $d(S_{n,k})+1$ when $\lfloor n/2\rfloor+1\le k\le n-2$ and n is even, or $d(S_{n,k})$ otherwise.

(2) If $p_1\ne1$, the lengths of paths are at most $d(S_{n,k})+2$.

Proof. If $p_1=1$ then $f=k-m-e-1\ge1$, such that $\Delta_1\ge2$, $\Delta_2\ge1$, and $\Delta_3\ge2$ by Lemma 2, Lemma 3 and Corollary 4. Notably, $l(\pi(\beta_i))$ can be computed as the α-function; however, 2 extra edges are added for moving β_i to position 1 at beginning of the construction and to its desired position at the last correction. Obviously, $l(\pi(\beta_i))$ is at most $d(P, \varepsilon)+2=d(S_{n,k})-\Delta_2+2\le d(S_{n,k})+1$ if $\lfloor n/2\rfloor+1\le k\le n-2$ and n is even or $d(P, \varepsilon)+2=d(S_{n,k})-\Delta_j+2\le d(S_{n,k})$, $j\in\{1, 3\}$ otherwise.

If $p_1\ne1$, then $f=k-m-e\ge1$, such that Δ_1, Δ_2, and $\Delta_3\ge2$, or $\Delta\ge2$ by Lemma 2, Lemma 3, and Corollary 4. Additionally, s_{1,k_1} (symbol 1) is corrected before other symbols in C_1, and extra 4 edges should be added for moving β_i to position 1 at beginning of the construction and to its desired position at the last correction, and R2, instead of R1, is applied to correct $s_{1,1}$ since s_{1,k_1} is just moved to position 1 before this correction. Therefore, $l(\pi(\beta_i))$ is at most $d(P, \varepsilon)+4= d(S_{n,k})-\Delta+4\le d(S_{n,k})+2$.

γ-function:

There are $e'=n-k-e$ outer external symbols of P. Recall that an outer external symbol of a vertex P is denoted by γ_i, where $1\le i\le e'$. The γ-function is designed to first replace p_1 with a γ_i, this can hop one step to P's neighbor, denoted by P_{γ_i}, and then keep γ_i in the label of each vertex of the constructed path $\pi(\gamma_i)$ until the other symbols have been corrected. Note that no $\pi(\gamma_i)$s can be constructed if $e'=0$.

If $p_1=1$, a $\pi(\gamma_i)$ can be constructed as follows.

Step 1. Replace p_1 with a γ_i; this can hop one step to P_{γ_i}, and its primary cycle is $(\gamma_i\ 1)$.

Step 2. Construct a path from P_{γ_i} to ε using the sequence $<C_1, C_2, ..., C_{c-1}, s_{c,k_c}, s_{c,1}, s_{c,2}, ..., s_{c,k_c-1}, \gamma_i.>$. Notably, if an external cycle exists then it must be C_c, and s_{c,k_c} is not a desired symbol such that γ_i is in the label of each vertex of $\pi(\gamma_i)$ until the other symbols have been corrected.

If $p_1\ne1$, a modified sequence is provided for keeping the correction of γ_i last, and a $\pi(\gamma_i)$ can be constructed as follows.

Step 1. Move γ_i to position 1. This movement can hop one step to P_{γ_i}, and its primary cycle is $(\gamma_i\ C_1)$.

Step 2. If s_{1,k_1} is not a desired symbol, construct a path from P_{γ_i} to ε using the sequence $<C_2, ..., C_c, s_{1,k_1}, s_{1,1}, s_{1,2}, ..., s_{1,k_1-1}, \gamma_i>$. If s_{1,k_1} is a desired symbol, construct a path from P_{γ_i} to ε using sequence $<C_2, ..., C_c, s_{1,k_1-1}, s_{1,1}, ..., s_{1,k_1-2}, \gamma_i, s_{1,k_1}>$, in which s_{1,k_1} is the last symbol, indicating that γ_i is to be replaced by the desired symbol s_{1,k_1} (=1) at the last correction.

Lemma 9. There are $n-k-e$ vertex-disjoint paths constructed using the γ-function; $\pi(\gamma_i)$ is vertex-disjoint to all $\pi(s_\alpha)$s and $\pi(\beta_i)$s.

Proof. There are $n-k$ external symbols, in which e inner external symbols are included in the label of P, thus, $e'=n-k-e$ outer external symbols can be used to construct path(s) by the γ-function. Clearly, every γ_i is distinct and all $\pi(\gamma_i)$s are vertex-disjoint. Since all vertices in a constructed path other than $\pi(\gamma_i)$ have γ_i at the desired position, each $\pi(\gamma_i)$ is vertex-disjoint to all $\pi(s_\alpha)$s and $\pi(\beta_i)$s.

Lemma 10. The length of each path constructed by the γ-function is at most $d(S_{n,k})+2$.

Proof. When $p_1=1$, by Lemma 2, Lemma 3 and Corollary 4, Δ_1, $\Delta_2\geq0$ and $\Delta_3\geq1$ since $e'\geq1$. Therefore, $l(\pi(\gamma_i))$ is at most $d(P, \varepsilon)+2= d(S_{n,k})-\Delta_3+2\leq d(S_{n,k})+1$ if $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is odd or $d(P, \varepsilon)+2=d(S_{n,k})-\Delta_j+2\leq d(S_{n,k})+2$, $j\in\{1, 2\}$ otherwise.

When $p_1\neq1$, the differences (Δ_1, Δ_2 and Δ_3) must be measured precisely. If $s_{1,1}$ is external, moving γ_i to position 1 can replace $s_{1,1}$, such that correcting $s_{1,1}$ is no longer required. Additionally, if C_1 is an internal cycle and $e>0$ (CEC exists), then moving γ_i to the position 1 causes C_1 to become external and can be combined with CEC. Notably, these two cases are mutually exclusive. Both cases reduce path length by 1. Therefore, the lengths of constructed paths are computed in the following situations.

(1) Symbol s_{1,k_1} is not desired. An additional 4 edges should be added for moving γ_i to position 1 at beginning of the construction and to its desired position last, moving s_{1,k_1} to position 1 and applying R2 instead of R1 to correct $s_{1,1}$. Therefore, $l(\pi(\gamma_i))\leq d(P, \varepsilon)+4$.

(1.1) Symbol $s_{1,1}$ is external. $l(\pi(\gamma_i))\leq d(P, \varepsilon)+3$ since the correction of $s_{1,1}$ is not needed. Thus, C_1 contains at least 3 symbols $s_{1,1}$, d_1, and s_{1,k_1}, such that $m'\geq1$ and $m+m'\geq2$. By Corollary 4 and $e'>0$, $m'+e'\geq2$ (≥3) for $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is even (odd). Lemmas 2 and 3 yield Δ_1, $\Delta_2\geq1$, and $\Delta_3\geq2$. Therefore, $l(\pi(\gamma_i)) = d(S_{n,k})-\Delta_3+3 \leq d(S_{n,k})+1$ if $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is odd, or $l(\pi(\gamma_i))=d(S_{n,k})-\Delta_j+3\leq d(S_{n,k})+2$, $j\in\{1, 2\}$ otherwise.

(1.2) Cycle C_1 is internal and $e>0$. $l(\pi(\gamma_i))\leq d(P, \varepsilon)+3$ because path length is reduced by 1. As C_1 is internal, $m\geq2$ and $m+m'\geq2$. By Corollary 4 and $e'>0$, $m'+e'\geq2$ (≥1) for $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is even (odd). Lemmas 2 and 3 yield Δ_1, Δ_2 and $\Delta_3\geq1$, or $\Delta\geq1$. Hence, $l(\pi(\gamma_i)) = d(S_{n,k})-\Delta+3 \leq d(S_{n,k})+2$.

(1.3) Otherwise: If $e=0$ and $e'\geq2$ since $k\leq n-2$, and at least one internal cycle exists such that $m\geq2$ and by Corollary 4, $m'+e'\geq2$ (≥3) for $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is even (odd). Lemma 3 acquires Δ_1, $\Delta_2\geq2$ and $\Delta_3\geq3$. Thus, $l(\pi(\gamma_i)) = d(S_{n,k})-\Delta_3+4 \leq d(S_{n,k})+1$ if $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is odd, or $l(\pi(\gamma_i))=d(S_{n,k})-\Delta_j+4\leq d(S_{n,k})+2$, $j\in\{1, 2\}$ otherwise. If $e>0$, C_1 is external and a CEC. Since $s_{1,1}$ must be internal, C_1 contains at least 4 symbols $s_{1,1}$, e_1, d_1, and s_{1,k_1}, such that $m'\geq2$ and $m+m'\geq4$. By Corollary 4, $m'+e'\geq4$ (≥3) for $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is even (odd). Lemma 2 yields Δ_1, Δ_2 and $\Delta_3\geq2$, or $\Delta\geq2$. Hence, $l(\pi(\gamma_i))=d(S_{n,k})-\Delta+4\leq d(S_{n,k})+2$.

(2) Symbol s_{1,k_1} is desired. Thus, symbol s_{1,k_1-1} is external. If C_2, ..., C_c are already corrected, γ_i will return to the position 1 automatically. Thus, an additional 3 edges should be added for moving γ_i to position 1 at beginning of the construction and to its desired position last; s_{1,k_1-1} and γ_i should be exchanged last to keep the correction of γ_i last. Therefore, $l(\pi(\gamma_i))\leq d(P, \varepsilon)+3$.

(2.1) Symbol $s_{1,1}$ is external. $l(\pi(\gamma_i))\leq d(P, \varepsilon)+2$ since the correction of $s_{1,1}$ is no longer required. By Corollary 4 and $e'>0$, $m'+e'\geq2$ (≥1) for $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is even (odd). Lemmas 2 and 3 yield $\Delta_1\geq0$, Δ_2 and $\Delta_3\geq1$. Thus, $l(\pi(\gamma_i)) = d(S_{n,k})-\Delta_1+2 \leq d(S_{n,k})+2$ if $2\leq k\leq\lfloor n/2\rfloor$, or $l(\pi(\gamma_i)) = d(S_{n,k})-\Delta_j+2 \leq d(S_{n,k})+1$, $j\in\{2, 3\}$ otherwise.

(2.2) Symbol $s_{1,1}$ is internal. Cycle C_1 contains at least 3 symbols $s_{1,1}$, e_1, and s_{1,k_1}, such that $m'\geq1$ and $m+m'\geq2$. By Corollary 4 and $e'>0$, $m'+e'\geq2$ (≥3) for $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is even (odd). Lemmas 2 and 3 yield Δ_1, $\Delta_2\geq1$ and $\Delta_3\geq2$. Thus, $l(\pi(\gamma_i))= d(S_{n,k})-\Delta_3+3\leq d(S_{n,k})+1$ if $\lfloor n/2\rfloor+1\leq k\leq n-2$ and n is odd, or $l(\pi(\gamma_i))=d(S_{n,k})-\Delta_j+3\leq d(S_{n,k})+2$, $j\in\{1, 2\}$ otherwise.

Lemmas 6, 7, 8, 9, 10 and 11 indicate that the $l(C_{n-1}(P, \varepsilon))$ is at most $d(S_{n,k})+2$, and can serve as an upper bound of $d_{n-1}(S_{n,k})$, as shown in Lemma

11.

Lemma 11. $d_{n-1}(S_{n,k}) \le d(S_{n,k})+2$.

4.2 Lower bound of $d_{n-1}(S_{n,k})$

An $S_{n,k}$ is a regular graph and $\kappa \ge 2$ if $2 \le k \le n-2$ [5]. The lower bound of $d_{n-1}(S_{n,k})$ is revealed by the following lemmas.

Lemma 12. If G is a regular graph with connectivity $\kappa \ge 2$, then $d_\kappa(G) \ge d(G)+1$.
Proof. Let x and $y \in V(G)$ be vertices with $d(x, y)=d(G)$. Because $\kappa \ge 2$, a vertex $z \ne y$ exists and is adjacent to x. One path of $C^*_\kappa(z, y)$ should be constructed from z via its neighbor x and then extended to y; the length of the constructed path is at least $d(x, y)+1=d(G)+1$. Therefore, $d_\kappa(G) \ge d(G)+1$.

Lemma 13. $d_{n-1}(S_{n,k}) \ge d(S_{n,k})+2$, where $2 \le k \le \lfloor n/2 \rfloor$.
Proof. Assuming $P=1e_2...e_k$ and $P_{e_1}=e_1e_2...e_k$ is one of its neighbor. The vertices $ye_2...e_k$, for y=2, ..., k, are common neighbors of P and P_{e_1}, and each should be included in a path of $C^*_{n-1}(P, \varepsilon)$ and such common neighbors are not contained in the path from P via P_{e_1} to ε. Thus, only $u=e_ie_2...e_{i-1}e_1e_{i+1}...e_k$ can serve as the successor vertex of P_{e_1} in the path. Since $d(u, \varepsilon)=d(S_{n,k})$, the path from P via P_{e_1} and then from u to ε has the length $d(S_{n,k})+2$. Thus, $l(C^*_{n-1}(P, \varepsilon))$ is $d(S_{n,k})+2$. Therefore, $d_{n-1}(S_{n,k}) \ge d(S_{n,k})+2$, where $2 \le k \le \lfloor n/2 \rfloor$.

According to Lemmas 11, 12 and 13, the following theorem is derived.

Theorem 14. $d_{n-1}(S_{n,k}) = d(S_{n,k})+2$, where $2 \le k \le \lfloor n/2 \rfloor$, and $d(S_{n,k})+1 \le d_{n-1}(S_{n,k}) \le d(S_{n,k})+2$, where $\lfloor n/2 \rfloor+1 \le k \le n-2$.

5 Conclusion

This work presented a novel general routing algorithm to construct a $C_{n-1}(P, \varepsilon)$ in an $S_{n,k}$ for each vertex P. This work also demonstrated that the length of each constructed $C_{n-1}(P, \varepsilon)$ is at most $d(S_{n,k})+2$, and $d_{n-1}(S_{n,k})$ is at least $d(S_{n,k})+2$ for $2 \le k \le \lfloor \le n/2 \rfloor$ or $d(S_{n,k})+1$ for $\lfloor n/2 \rfloor+1 \le k \le n-2$. That is, the constructed $C_{n-1}(P, \varepsilon)$ using the proposed algorithm is best if $2 \le k \le \lfloor n/2 \rfloor$ and nearly best if $\lfloor n/2 \rfloor+1 \le k \le n-2$. These measurement results demonstrate that $d_{n-1}(S_{n,k})$ is $d(S_{n,k})+2$ if $2 \le k \le \lfloor \le n/2 \rfloor$, or between $d(S_{n,k})+1$ and $d(S_{n,k})+2$ if $\lfloor n/2 \rfloor+1 \le k \le n-2$. Actually, given any vertex P in some $S_{n,k}$s with $\lfloor n/2 \rfloor+1 \le k \le n-2$, a $C_{n-1}(P, \varepsilon)$ with a length at most $d(S_{n,k})+1$ can be built by modifying the $C_{n-1}(P, \varepsilon)$ constructed by the proposed algorithm. Therefore, we claim that the wide diameters of some $S_{n,k}$s are $d(S_{n,k})+1$ for $\lfloor n/2 \rfloor+1 \le k \le n-2$. Future work will shorten some constructed $C_{n-1}(P, \varepsilon)$s of length $d(S_{n,k})+2$ for $\lfloor n/2 \rfloor+1 \le k \le n-2$ by reconstructing each of the longest paths.

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